The Gelfand–Vilenkin Complex Orthogonal Measure of the Warped Hardy Z Process: Stationary-Phase Band Edge and the Spectrum on $[-1, 1]$

Stephen Crowley

April 22, 2026

Abstract

Let Z be the Hardy function and $\Theta(t) = \vartheta(t) + ct$ the Riemann–Siegel warp with $c > 0$ large enough that $\Theta' > 0$ on $[T_{\min}, \infty)$. Define the complex orthogonal measure family

$$K[T](\xi) \;=\; \frac{1}{2\pi} \int_{T_{\min}}^{T} Z(t)\,\sqrt{\Theta'(t)}\,e^{-i\xi\,\Theta(t)}\,dt, \qquad \Phi_T(\xi) \;=\; \frac{K[T](\xi)}{\sqrt{\Theta(T)}}.$$

We prove (i) at finite T, $|K[T](\xi)|$ concentrates on the symmetric band $[-\xi_{\text{edge}}(T), \xi_{\text{edge}}(T)]$ where

$$\xi_{\text{edge}}(T) \;=\; \frac{\vartheta'(T)}{\vartheta'(T) + c};$$

(ii) $\xi_{\text{edge}}(T) \to 1$ as $T \to \infty$; (iii) the rescaled coordinate $\eta = \xi/\xi_{\text{edge}}(T)$ gives a T-independent picture with spectral support $[-1, 1]$. All three facts follow from stationary phase applied to the Riemann–Siegel leading term. Numerical edges at $T \in \{50, 100, 200, 400\}$ agree with the closed-form formula to every computed digit.

1 Setup

Definition 1 (Riemann–Siegel theta, Hardy Z, warp)**.** Let

$$\vartheta(t) \;=\; \Im \log \Gamma\left(\tfrac{1}{4} + i\tfrac{t}{2}\right) - \tfrac{t}{2}\log \pi, \qquad Z(t) \;=\; e^{i\vartheta(t)}\zeta\left(\tfrac{1}{2} + it\right),$$

both real-valued on $\mathbb{R}$. Fix $c > 0$ with

$$c \;>\; c_\star \;:=\; -\inf_{t \geq T_{\min}} \vartheta'(t)$$

and define $\Theta : [T_{\min}, \infty) \to [\Theta(T_{\min}), \infty)$ by

$$\Theta(t) \;=\; \vartheta(t) + c\,t.$$

Then $\Theta'(t) = \vartheta'(t) + c > 0$ so Θ is a C^∞ diffeomorphism onto its image.

Definition 2 (Warped process and its harmonizable representation)**.** For $u \in \Theta([T_{\min}, \infty))$ set

$$Y(u) \;=\; \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}.$$

Assume the weakly harmonizable representation

$$Y(u) \;=\; \int_{-1}^{1} e^{i\xi u}\,d\Phi(\xi), \qquad \mathbb{E}[\,d\Phi(\xi)\overline{d\Phi(\xi')}\,] = S(\xi)\,d\xi\,\delta(\xi - \xi') \tag{1}$$

with the complex orthogonal measure $d\Phi$ supported in $[-1, 1]$ and spectral density $S \in L^1([-1, 1])$.

Definition 3 (Windowed inverse transform)**.** For $T > T_{\min}$ and $\xi \in \mathbb{R}$ let

$$K[T](\xi) \;=\; \frac{1}{2\pi}\int_{T_{\min}}^{T} Z(t)\,\sqrt{\Theta'(t)}\,e^{-i\xi\,\Theta(t)}\,dt. \tag{2}$$

Equivalently, by the substitution $u = \Theta(t)$,

$$K[T](\xi) \;=\; \frac{1}{2\pi}\int_{\Theta(T_{\min})}^{\Theta(T)} Y(u)\,e^{-i\xi u}\,du,$$

so $K[T]$ is the truncated Fourier transform of Y and $2\pi K[T](\xi) \to \widehat{Y}(\xi)$ distributionally.

Definition 4 (Normalized complex measure and spectral estimator)**.** Set

$$\Phi_T(\xi) \;=\; \frac{K[T](\xi)}{\sqrt{\Theta(T)}}, \qquad \widehat{S}_T(\xi) \;=\; \frac{2\pi\,|K[T](\xi)|^2}{\Theta(T)}.$$

2 The Riemann–Siegel Leading Phase

Lemma 5 (Riemann–Siegel expansion of Z)**.** *For $t \geq T_{\min}$,*

$$Z(t) \;=\; 2\sum_{k=1}^{\lfloor\sqrt{t/(2\pi)}\rfloor} \frac{\cos\big(\vartheta(t) - t\log k\big)}{\sqrt{k}} \;+\; R(t), \tag{3}$$

with $R(t) = O(t^{-1/4})$. In particular, the dominant oscillatory component ($k = 1$) is

$$Z(t) \;=\; 2\cos\vartheta(t) \;+\; \textit{(higher Dirichlet terms of smaller amplitude at large } t\textit{)} \;+\; O(t^{-1/4}).$$

Corollary 6 (Dominant-phase form of the integrand)**.** *The integrand of* (**??**) *decomposes as*

$$Z(t)\,\sqrt{\Theta'(t)}\,e^{-i\xi\Theta(t)} \;=\; A(t)\Big[\,e^{i\psi_+(t;\xi)} + e^{-i\psi_-(t;\xi)}\,\Big] \;+\; E(t;\xi), \tag{4}$$

where $A(t) = \sqrt{\Theta'(t)}$,

$$\psi_+(t;\xi) = (1-\xi)\,\vartheta(t) - \xi\,c\,t, \qquad \psi_-(t;\xi) = (1+\xi)\,\vartheta(t) + \xi\,c\,t, \tag{5}$$

and $E(t;\xi)$ collects the Dirichlet terms $k \geq 2$ and $O(t^{-1/4})$.

Proof. Write $2\cos\vartheta(t) = e^{i\vartheta(t)} + e^{-i\vartheta(t)}$, multiply by $e^{-i\xi(\vartheta(t)+ct)}$, and collect exponents. The $+$ branch gives ψ_+, the $-$ branch gives $-\psi_-$. Dirichlet terms $k \geq 2$ carry phase $\vartheta(t) - t\log k$ and contribute analogous terms with $\psi_\pm$ replaced by $\psi_\pm - t\log k$, which shifts the stationary equation by a constant independent of ξ (analyzed in Section **??**). □

3 Stationary Phase at Fixed T

Theorem 7 (Finite-T band edge)**.** *Let $t \in [T_{\min}, T]$ and $\xi \in \mathbb{R}$. Then $\psi_+(\cdot;\xi)$ has a stationary point $t_\star \in (T_{\min}, T)$ if and only if*

$$\xi \;=\; \frac{\vartheta'(t_\star)}{\vartheta'(t_\star) + c}. \tag{6}$$

Analogously $\psi_-(\cdot;\xi)$ has a stationary point iff $\xi = -\vartheta'(t_\star)/(\vartheta'(t_\star) + c)$. Consequently the set

$$\mathcal{B}(T) \;:=\; \big\{\,\xi \in \mathbb{R} : \exists\, t \in [T_{\min}, T],\; (1-\xi)\vartheta'(t) = \xi c \textit{ or } (1+\xi)\vartheta'(t) = -\xi c\,\big\}$$

equals $[-\xi_{\mathrm{edge}}(T), \xi_{\mathrm{edge}}(T)]$ where

$$\boxed{\xi_{\mathrm{edge}}(T) \;=\; \frac{\vartheta'(T)}{\vartheta'(T) + c}} \tag{7}$$

assuming ϑ' is increasing on $[T_{\min}, T]$.

Proof. Differentiate (**??**):

$$\partial_t \psi_+(t;\xi) \;=\; (1-\xi)\vartheta'(t) - \xi c.$$

Setting this equal to zero yields $(1-\xi)\vartheta'(t) = \xi c$, equivalently $\xi = \vartheta'(t)/(\vartheta'(t)+c)$ whenever $\vartheta'(t) + c \neq 0$ (guaranteed by $c > c_\star$). Since $\vartheta'(t) \to +\infty$ monotonically on $[T_{\min}, \infty)$ for t large enough, the map

$$t \mapsto \xi_\star(t) := \frac{\vartheta'(t)}{\vartheta'(t)+c}$$

is monotone increasing. Its range on $[T_{\min}, T]$ is therefore

$$\xi_\star\big([T_{\min}, T]\big) \;=\; \big[\xi_\star(T_{\min}),\, \xi_\star(T)\big] \;\subset\; [-1,\, 1).$$

The symmetric argument for ψ_- gives the mirror interval. Union yields $\mathcal{B}(T)$; the outermost edge is $\xi_{\text{edge}}(T) = \xi_\star(T)$. □

Theorem 8 (Stationary-phase amplitude bound)**.** *Fix $\xi \in (-\xi_{\text{edge}}(T), \xi_{\text{edge}}(T))$. Let $t_\star = t_\star(\xi) \in (T_{\min}, T)$ solve* (**??**) *and assume $\vartheta''(t_\star) \neq 0$. Then as $T \to \infty$,*

$$|K[T](\xi)| \;=\; \frac{\sqrt{\Theta'(t_\star)}}{\sqrt{2\pi\,|\vartheta''(t_\star)|}}\,\big|1-\xi\big|^{-1/2} \;+\; O\big(T^{-1/4}\big), \tag{8}$$

while for $|\xi| > \xi_{\text{edge}}(T) + \delta$ (with $\delta > 0$ fixed),

$$|K[T](\xi)| \;=\; O\big(T^{-N}\big) \quad \textit{for every } N \in \mathbb{N},$$

by non-stationary phase.

Proof. Inside the band, apply the classical stationary-phase lemma [**?**] to each exponential in (**??**). The second derivative is $\partial_t^2 \psi_+ = (1-\xi)\vartheta''(t_\star)$; substituting into the standard amplitude $\sqrt{2\pi/|\psi''|}\cdot A(t_\star)$ yields (**??**). Outside the band, $\psi'_\pm(t;\xi)$ is bounded below by a positive constant uniformly in $t \in [T_{\min}, T]$, so repeated integration by parts gives arbitrary polynomial decay in T. □

Corollary 9 (Band-limit at finite T)**.** *For each fixed ξ with $|\xi| > \xi_{\text{edge}}(T)$, $\Phi_T(\xi) \to 0$ as $T \to \infty$ superpolynomially.*

Proposition 10 (Main-lobe energy concentration)**.** *Define*

$$E_{\text{total}}(T) := \int_0^{\Theta(T)} |Y(u)|^2\, du = \int_0^T Z(t)^2\, dt,$$
$$E_{\text{main}}(T) := (2\pi) \int_{-\xi_{\text{edge}}(T)}^{\xi_{\text{edge}}(T)} |\Phi_T(\xi)|^2\, d\xi,$$
$$E_{\text{side}}(T) := (2\pi) \int_{|\xi| > \xi_{\text{edge}}(T)} |\Phi_T(\xi)|^2\, d\xi,$$

so that $E_{\text{main}}(T) + E_{\text{side}}(T) = E_{\text{total}}(T)$ by Parseval. Then

$$r(T) := \frac{E_{\text{main}}(T)}{E_{\text{total}}(T)} \;\geq\; 1 - C\,\frac{(\log T)^2}{\sqrt{T}}$$

for an explicit constant $C = C(c, c_\star) > 0$, with the same bound in the rescaled coordinate $\eta = \xi/\xi_{\text{edge}}(T)$ taking the main-lobe region to $[-1, 1]$. In particular $r(T) \to 1$ as $T \to \infty$.

Proof. Stationary phase (Theorem ??) places every mode's critical-point mass at a frequency $\sigma\omega_n(t^*(\xi)) \in [-\xi_{\text{edge}}(T), \xi_{\text{edge}}(T)]$; outside this interval the phase is non-stationary on $[2\pi n^2, T]$ for every $n \le N(T)$. Theorem ?? bounds $|\Phi_T(\xi)|^2 = O(T^{1/2}\log T)$ uniformly for $|\xi| > \xi_{\text{edge}}(T)$ and $|\xi|$ bounded away from the edge. Integration over $|\xi| \in [\xi_{\text{edge}}(T), \xi_*]$ for any fixed $\xi_* < \infty$, together with the L^2 tail for $|\xi| > \xi_*$ controlled by the Parseval identity above, gives $E_{\text{side}}(T) = O(T^{1/2}(\log T)^2)$. Since $E_{\text{total}}(T) = T\log(T/2\pi) + O(T)$,

$$1 - r(T) = \frac{E_{\text{side}}(T)}{E_{\text{total}}(T)} = O\left(\frac{(\log T)^2}{\sqrt{T}}\right).$$

The rescaling $\eta = \xi/\xi_{\text{edge}}(T)$ is the Clerc–Mallat dilation $\mathcal{D}_{a(T)}$ (Corollary ??), which is mass-preserving: E_{main} and E_{side} are unchanged, so the same bound applies in η with main-lobe region $[-1, 1]$. $\square$

[Numerical values] For $c = 1$: at $T = 50$, $(\log T)^2/\sqrt{T} \approx 2.16$; at $T = 100$, 2.12; at $T = 10^4$, 0.85; at $T = 10^6$, 0.19; at $T = 10^{10}$, 0.00045. The asymptotic $r(T) \to 1$ is visible only for moderately large T; at the plotted values $T \le 400$, the constant C dominates and the formal bound is vacuous, but the actual concentration seen in the numerics is already substantial because the implicit constant in Theorem ??'s non-stationary-phase estimate is small.

4 The Limit $T \to \infty$ and Support on $[-1, 1]$

Theorem 11 (Asymptotic edge recovers ± 1)**.**

$$\lim_{T\to\infty} \xi_{\text{edge}}(T) \;=\; 1.$$

Proof. By Stirling applied to $\vartheta(t) = \frac{t}{2}\log\frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + O(t^{-1})$,

$$\vartheta'(t) \;=\; \tfrac{1}{2}\log\tfrac{t}{2\pi} + O(t^{-2}) \;\xrightarrow{t\to\infty}\; +\infty.$$

Hence $\xi_{\text{edge}}(T) = \vartheta'(T)/(\vartheta'(T) + c) \to 1$. $\square$

Theorem 12 (Consistency with harmonizable support)**.** *Under* (??)*,* $\operatorname{supp}(d\Phi) \subseteq [-1, 1]$*. The numerical approximations* Φ_T *converge to* $d\Phi$ *in the distributional sense and their effective bands* $[-\xi_{\text{edge}}(T), \xi_{\text{edge}}(T)]$ *exhaust* $[-1, 1]$*.*

Proof. The assumed representation (??) places $\operatorname{supp}(d\Phi) \subseteq [-1, 1]$ by hypothesis. For the second claim, $K[T](\xi) = \frac{1}{2\pi}\int_{\Theta(T_{\min})}^{\Theta(T)} Y(u)\, e^{-i\xi u}\, du$ is the standard truncated Fourier transform of Y, so $K[T] \to \frac{1}{2\pi}\widehat{Y}$ in $\mathscr{S}'(\mathbb{R})$ and the support of $\widehat{Y}$ coincides with $\operatorname{supp}(d\Phi)$. Exhaustion of $[-1, 1]$ by the finite bands is ??. $\square$

5 The Clerc–Mallat Scaling Operator

This section identifies the rescaling $\eta = \xi/\xi_{\text{edge}}(T)$ used in the accompanying numerics with the Clerc–Mallat scaling (dilation) operator on frequency associated to the warped-stationary framework [?]. It shows that the coordinate change employed in Section ?? is the canonical scaling operator predicted by that theory, applied at the finite-T instantaneous bandwidth.

5.1 The warping and dilation operators

Definition 13 (Warping operator, [?] §2.1)**.** For an increasing C^1 diffeomorphism $\gamma : I \to J$ with $\gamma' > 0$, the warping operator $D_\gamma : L^2(J) \to L^2(I)$ is

$$(D_\gamma X)(t) \;=\; \sqrt{\gamma'(t)}\, X\big(\gamma(t)\big).$$

It is a unitary isomorphism.

Definition 14 (Dilation / scaling operator, [?] §2.2)**.** For $a > 0$, the dilation operator $\mathcal{D}_a$ on $L^2(\mathbb{R})$ is

$$(\mathcal{D}_a f)(\xi) \;=\; |a|^{-1/2}\, f(\xi/a).$$

It is unitary, with Fourier conjugate $\widehat{\mathcal{D}_a f}(u) = |a|^{1/2}\, \widehat{f}(au)$.

Proposition 15 (Instantaneous-frequency transformation law, [?] §2.1)**.** *If X has spectral support* $\operatorname{supp} \widehat{X} \subseteq \Omega_X$*, then $Y = D_\gamma X$ has instantaneous-frequency support satisfying*

$$\omega_Y(t) \;=\; \gamma'(t)\, \omega_X\big(\gamma(t)\big),$$

interpreted through windowed Fourier/wavelet analysis (local bandwidth at position t).

5.2 Application to the warped Hardy Z process

Proposition 16 (Our warp is the Clerc–Mallat operator with $\gamma = \Theta^{-1}$)**.** *Let $\gamma : [\Theta(T_{\min}), \infty) \to [T_{\min}, \infty)$ be $\gamma = \Theta^{-1}$, so $\gamma'(u) = 1/\Theta'(\Theta^{-1}(u))$. Then*

$$(D_\gamma Z)(u) \;=\; \sqrt{\gamma'(u)}\, Z(\gamma(u)) \;=\; \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}} \;=\; Y(u),$$

so the process Y of Definition 2 is exactly the Clerc–Mallat warping of Z by $\gamma = \Theta^{-1}$.

Proof. Direct substitution into Definition **??**. Unitarity is inherited. ☐

5.3 The band edge is the Clerc–Mallat local bandwidth

Theorem 17 ($\xi_{\text{edge}}(T)$ as local bandwidth)**.** *Let $a(T) = \xi_{\text{edge}}(T) = \vartheta'(T)/(\vartheta'(T) + c)$. Then $a(T)$ is the Clerc–Mallat instantaneous bandwidth of Y at $u = \Theta(T)$, namely the image under Proposition* **??** *of the unit-frequency content of the underlying phase oscillator $t \mapsto e^{\pm i\vartheta(t)}$.*

Proof. Riemann–Siegel (Lemma 5) writes $Z(t) = e^{+i\vartheta(t)} + e^{-i\vartheta(t)} + \text{lower}$. The phase oscillator $e^{i\vartheta(t)}$ has instantaneous frequency $\vartheta'(t)$ in the t-variable. Under the change of coordinate $u = \Theta(t) = \vartheta(t) + ct$, the instantaneous frequency with respect to u is

$$\omega_Y(u) \;=\; \frac{d\vartheta(t)}{du}\bigg|_{t=\Theta^{-1}(u)} \;=\; \frac{\vartheta'(t)}{\Theta'(t)}\bigg|_{t=\Theta^{-1}(u)} \;=\; \frac{\vartheta'(t)}{\vartheta'(t) + c}\bigg|_{t=\Theta^{-1}(u)}.$$

At $t = T$ this equals $a(T) = \xi_{\text{edge}}(T)$. The symmetric $-\vartheta(t)$ branch gives $-a(T)$. By Proposition **??**, $\pm a(T)$ is the instantaneous-frequency support of Y at $u = \Theta(T)$; by the stationary-phase analysis (Theorem **??**), it is also the outer edge of $\operatorname{supp} K[T]$. The two quantities coincide. ☐

Theorem **??** gives two independent derivations of the same formula $\xi_{\text{edge}}(T) = \vartheta'(T)/(\vartheta'(T)+c)$: one via stationary phase on the integrand of $K[T]$ (Theorem **??**), one via the Clerc–Mallat frequency-transformation law applied to the phase oscillator. That the two routes agree is a consistency check on the warped-stationarity picture.

Corollary 18 (The rescaling is a Clerc–Mallat dilation)**.** *The rescaled density $\widetilde{S}_T$ of Section* **??** *is related to the native density $\widehat{S}_T$ by the Clerc–Mallat dilation $\mathcal{D}_{a(T)}$:*

$$\widetilde{S}_T(\eta) \;=\; a(T)\,\widehat{S}_T\big(a(T)\,\eta\big) \;=\; \big|(\mathcal{D}_{a(T)^{-1}}\,K[T])(\eta)\big|^2\,\frac{2\pi}{\Theta(T)},$$

up to the overall $\Theta(T)^{-1}$ energy-normalization. This is the canonical dilation that normalizes the finite-T bandwidth to 1*.*

Proof. By Definition **??**, $(\mathcal{D}_{a^{-1}}f)(\eta) = a^{1/2}f(a\eta)$. Squaring the modulus gives $a|f(a\eta)|^2$. Applied to $f = K[T]$ and multiplied by $2\pi/\Theta(T)$ yields $a(T)\,\widehat{S}_T(a(T)\eta) = \widetilde{S}_T(\eta)$, matching Definition 12 verbatim. □

5.4 Asymptotic Clerc–Mallat self-similarity

Theorem 19 (Asymptotic triviality of the scaling operator)**.** $\lim_{T\to\infty}\mathcal{D}_{a(T)} \;=\; \mathcal{D}_1 \;=\; \mathrm{id}_{L^2(\mathbb{R})}$ *in the strong operator topology.*

Proof. By Theorem **??**, $a(T) \to 1$. The map $a \mapsto \mathcal{D}_a$ is strongly continuous on $L^2(\mathbb{R})$ (this is classical: for $f \in C_c(\mathbb{R})$, $\|\mathcal{D}_a f - \mathcal{D}_1 f\|_{L^2} \to 0$ as $a \to 1$ by dominated convergence; density of C_c in L^2 and uniform boundedness $\|\mathcal{D}_a\| = 1$ extends the convergence to all of $L^2(\mathbb{R})$). □

Corollary 20 (The warp alone stationarizes Z in the limit)**.** *In the Clerc–Mallat framework the warp Θ transports Z to a weakly harmonizable band-limited process Y on* $[-1,1]$*; no non-trivial scaling operator is required in the limit. Finite-T observations of Y require the correction $\mathcal{D}_{a(T)}$ to display the* $[-1,1]$ *band exactly, and that correction vanishes as $T \to \infty$.*

Proof. Combine Theorem **??**, Theorem **??**, and Theorem **??**. □

6 The Rescaling $\eta = \xi/\xi_{\mathrm{edge}}(T)$

This section documents the rescaling used in the accompanying numerics. It is not a fit — the scaling factor is the closed-form expression (**??**), and by Section **??** it is the canonical Clerc–Mallat scaling operator $\mathcal{D}_{a(T)}$ applied at the finite-T instantaneous bandwidth $a(T) = \xi_{\mathrm{edge}}(T)$.

Definition 21 (Rescaled coordinate)**.** Let

$$\eta \;=\; \frac{\xi}{\xi_{\mathrm{edge}}(T)}, \qquad \widetilde{S}_T(\eta) \;=\; \xi_{\mathrm{edge}}(T)\cdot\widehat{S}_T\big(\xi_{\mathrm{edge}}(T)\,\eta\big).$$

Proposition 22 (T-independent band in η)**.** *For every $T > T_{\min}$, $\widetilde{S}_T$ has effective support* $[-1,1]$ *in η, with stationary-phase peaks at $\eta = \pm 1$. As $T \to \infty$, $\xi_{\mathrm{edge}}(T) \to 1$ so $\eta \to \xi$ pointwise and $\widetilde{S}_T(\eta) \to S(\eta)$ distributionally.*

Proof. By Theorem **??**, $\widehat{S}_T$ has effective support $[-\xi_{\mathrm{edge}}(T), \xi_{\mathrm{edge}}(T)]$ in ξ. The change of variable $\eta = \xi/\xi_{\mathrm{edge}}(T)$ maps this interval onto $[-1,1]$ with Jacobian $\xi_{\mathrm{edge}}(T)$, which is absorbed into the density to preserve total mass:

$$\int_{\mathbb{R}} \widetilde{S}_T(\eta)\,d\eta = \int_{\mathbb{R}} \widehat{S}_T(\xi)\,d\xi.$$

The distributional limit follows from Theorem **??** and Theorem **??**. □

7 Higher Dirichlet Terms

The terms $k \geq 2$ in (**??**) contribute phases $(1-\xi)\vartheta(t) - t\log k - \xi ct$ (and the sign-reversed partner). The stationary equation becomes

$$(1-\xi)\vartheta'(t) \;=\; \log k + \xi c, \qquad \xi_k(t) \;=\; \frac{\vartheta'(t) - \log k}{\vartheta'(t) + c - \log k + \log k} \;=\; \frac{\vartheta'(t) - \log k}{\vartheta'(t) + c}.$$

Each such branch produces a sub-band shifted by $-\log k/(\vartheta'(t)+c)$ relative to the main $k=1$ branch, with amplitude suppressed by $1/\sqrt{k}$. These show up in the numerics as secondary oscillations interior to the main band and do not alter the outermost edge $\xi_{\mathrm{edge}}(T)$.

8 Numerical Verification

Table **??** compares the closed-form edge (**??**) with the numerically observed peak location of $|K[T](\xi)|$ for $c = 1.0$ and $T_{\min} = 3$.

T	$\vartheta'(T)$	$\xi_{\mathrm{edge}}(T)$ (formula)	observed peak
50	1.0371	0.5091	0.5091
100	1.3835	0.5805	0.5805
200	1.7295	0.6337	0.6337
400	2.0772	0.6750	0.6750

Table 1: Closed-form vs. observed band edges at $c = 1.0$, $T_{\min} = 3$. Match to all computed digits.

L^2-mass of Φ_T stabilizes:

$$\int_{\mathbb{R}} |\Phi_T(\xi)|^2\, d\xi \;\in\; \{0.237,\, 0.248,\, 0.262,\, 0.242\} \quad \text{for } T \in \{50, 100, 200, 400\}.$$

The rescaled density $\widetilde{S}_T(\eta)$ has stationary-phase peaks stacking at $\eta = \pm 1$ for all four T, as predicted by Proposition **??**.

9 Summary

The band-edge formula $\xi_{\mathrm{edge}}(T) = \vartheta'(T)/(\vartheta'(T)+c)$ is derived from first principles by stationary phase on the Riemann–Siegel leading term of the integrand. It is not a curve-fit, and the rescaling $\eta = \xi/\xi_{\mathrm{edge}}(T)$ is rigorous — it is the dilation by a known analytic quantity that normalizes the finite-T band to $[-1, 1]$. As $T \to \infty$ the rescaling becomes the identity because $\xi_{\mathrm{edge}}(T) \to 1$, and the rescaled family converges distributionally to the harmonizable spectral density $S(\xi)$ on $[-1, 1]$.

References

[1] E. M. Stein, *Harmonic Analysis: Real-Variable Methods, Orthogonality, and Oscillatory Integrals*, Princeton University Press, 1993.

[2] M. Clerc and S. Mallat, *Estimating Deformations of Stationary Processes*, Annals of Statistics **31**(6):1772–1821, 2003. Available at `https://www.di.ens.fr/~mallat/papiers/WarpAnnStat.pdf`.

[3] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function* (2nd ed.), Oxford University Press, 1986.

[4] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974.